MILITARY SPECIFICATION

ATTENUATORS, FIBER OPTIC, SHIPBOARD GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for fiber optic attenuators (see 6.6.2) intended for Naval shipboard applications. Attenuators covered by this specification are single channel passive devices that cause a known level of reduction in optical power by non-electrical means.

1.2 Classification. Attenuators covered by this specification are classified as specified in 1.2.1 through 1.2.3.

1.2.1 Type. The type designation defines whether the level of optical power reduction is fixed or variable as follows:

Type F - Fixed
Type V - Variable

1.2.2 Port construction. The port construction designation defines the method of interfacing attenuators to the fiber optic cables as follows:

Port construction A - Single-mode pigtail
Port construction B - Multimode pigtail
Port construction C - Connectorized, single-mode
Port construction D - Connectorized, multimode

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.3 Package. The package designation defines the type of packaging as follows:

Package S - Stand-alone (SA) package
Package P - Printed circuit board (PCB) mounted

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-P-35	- Passivation Treatments for Corrosion-Resistant Steel.
TT-I-735	- Isopropyl Alcohol.

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MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8 ST.
MIL-F-16884	- Fuel, Navy Distillate.
MIL-L-17331	- Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-L-23699	- Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-C-55330	- Connectors, Electrical and Fiber Optic, Packaging of.
MIL-C-85045	- Cable, Fiber Optic, Shipboard (Metric) General Specification for.
MIL-A-24726/1	- Attenuator, Fiber Optic, Shipboard, Fixed, Connectorized, Single-Mode, Stand-Alone.

STANDARDS

FEDERAL

FED-STD-313	- Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

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MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.
MIL-STD-1344 - Test Methods for Electrical Connectors.
DOD-STD-1678 - Fiber Optics Test Methods and Instrumentation.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

455 - Standard Test Procedures for Fiber Optic Fibers, Cables, Transducers, Connecting and Terminating Devices.
455-2 - Impact Test Measurements for Fiber Optic Devices. (DoD adopted)
455-4 - Fiber Optic Connector/Component Temperature Life. (DoD adopted)
455-5 - Humidity Test Procedure for Fiber Optic Connecting Devices. (DoD adopted)
455-6 - Cable Retention Test Procedure for Fiber Optic Cable Interconnecting Devices. (DoD adopted)
455-12 - Fluid Immersion Test Procedure for Fiber Optic Connecting Devices.
455-13 - Visual and Mechanical Inspection of Fibers, Cables, Connectors and/or Other Fiber Optic Devices. (DoD adopted)
455-16 - Salt Spray (Corrosion) Test for Fiber Optic Components.
455-20 - Measurement of Change in Optical Transmittance. (DoD adopted)
455-21 - Mating Durability for Fiber Optic Interconnecting Devices. (DoD adopted)
455-22 - Ambient Light Susceptibility of Fiber Optic Components. (DoD adopted)
455-34 - Interconnection Device Insertion Loss Test. (DoD adopted)
455-35 - Fiber Optic Connector Dust (Fine Sand) Test.
455-36 - Twist Test for Fiber Optic Cable Assemblies. (DoD adopted)
455-49 - Procedure for Measure Nuclear Radiation Effects in Fiber Optic Components.
455-71 - Procedure to Measure Temperature Shock Effects on Fiber Optic Components.

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(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.3 Materials. The materials selected shall have an intended lifetime of 20 years. They shall be of lightest practical material suitable for its intended purpose. All combinations of materials shall be non-magnetic and nonnutrient. The materials shall have no adverse effect on the health of personnel when used for its intended purpose. All specified materials are subject to toxicological data and formulations review and/or inspection for safety of the material by the Government. Questions concerning health effects shall be referred by the contracting activity to the departmental medical service who will act as an advisor to the contracting activity.

3.3.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new. Products may be fabricated using raw materials produced from recovered bulk materials to the extent practicable if the intended use of the product is not jeopardized. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become part of a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of partially processed, assembled, used or rebuilt products are allowed under this specification.

3.3.2 Metals. Metals shall be corrosion resistant and shall have a passivated surface that is compatible with external coatings or platings of the type and color specified in the specification sheets (see 3.1).

3.3.3 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless finished to prevent electrolytic corrosion. Selection and protection of dissimilar metal combinations shall be in accordance with MIL-STD-889.

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3.3.4 Finish. The resultant finish on all attenuators shall be:

(a) Stainless steel components - passivate in accordance with QQ-P-35.

3.3.5 Nonmetallic materials. Nonmetallic materials used in the construction of attenuators shall not be degraded under the specified environmental conditions.

3.3.6 Material safety data sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.7).

3.4 Design and construction. The configuration and physical dimensions of the attenuator shall be as specified in the specification sheets (see 3.1). The attenuator shall consist of an assembly with two optical ports such that either port may serve as the input port while the other serves as the output port. The external port construction shall consist of either connectors or pigtails as the interface to the fiber optic cable. Attenuators with the same part number shall be physically and functionally interchangeable.

3.4.1 Variable attenuators. Type V attenuators shall have a locking mechanism to prevent a change in attenuation over time.

3.4.2 Fiber optic pigtail. Attenuators with port construction A or P shall be supplied with two fiber or cable pigtails as specified in the specification sheets (see 3.1). One end of the pigtail shall be permanently mounted in the attenuator housing, and the mounting construction shall provide strain relief for the pigtail. The pigtail length shall be not less than 1 meter. No metal components shall be used within the pigtail, except for mounting to the attenuator housing or termination connector.

3.4.3 Fiber optic connectors. Attenuators with port construction C shall be supplied with fiber optic connectors as specified in the specification sheet (see 3.1).

3.4.4 Operating hardware (for type V attenuators only). The mechanism for adjusting the optical attenuation for type V attenuators shall be provided as specified in the specification sheet (see 3.1).

3.4.5 Mounting hardware. The attenuator shall be provided with mounting hardware as specified in the specification sheet (see 3.1).

3.4.6 Seals. Grommets, O-rings, interfacial seals, boots, gaskets, or any other combination of sealing devices, as needed by the particular attenuator design, shall be provided as specified in the specification sheet (see 3.1).

3.4.7 Enclosures. As specified in the specification sheet (see 3.1), the type of enclosure shall be either hermetic or watertight (see 3.5.3.5 and 3.5.3.6).

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3.4.8 Finish (see 4.6.1.1). The attenuator finish shall have no peeling or chipping of the plating or finishes, nicks, burrs, or other surface blemishes.

3.4.9 Size (see 4.6.1.1). The size of the attenuator shall be as specified in the specification sheet (see 3.1).

3.4.10 Weight (see 4.6.2.1). The weight of the attenuator shall be as specified in the specification sheet (see 3.1).

3.4.11 Color (see 4.6.1.1). The color of the attenuator shall be as specified in the specification sheet (see 3.1).

3.4.12 Maintainability. The attenuator shall require no preventive maintenance.

3.5 Performance requirements. The performance requirements shall be defined in terms of optical, mechanical, environmental, and chemical properties.

3.5.1 Optical properties.

3.5.1.1 Fixed value attenuators.

3.5.1.1.1 Insertion loss (see 4.6.3.1). The insertion loss (see 6.6.3) between the input port and the output port shall be within the tolerance specified in the specification sheet (see 3.1).

3.5.1.2 Variable attenuators.

3.5.1.2.1 Insertion loss (see 4.6.3.1). The insertion loss between the input port and the output port shall be within the tolerance as specified in the specification sheet (see 3.1).

3.5.1.2.2 Dynamic range (see 4.6.3.2). The maximum and minimum levels of insertion loss shall be as specified in the specification sheet (see 3.1).

3.5.1.2.3 Resolution (see 4.6.3.3). The resolution of the insertion loss of the attenuator and attenuator steps shall be as specified in the specification sheet (see 3.1).

3.5.1.3 Change in optical transmittance (see 4.6.3.4). The change in optical transmittance shall be not greater than 0.5 decibel (dB) for any environmental or mechanical requirement.

3.5.1.4 Return loss (see 4.6.3.5). The return loss (see 6.6.4) shall be not greater than minus 26 dB for single-mode attenuators.

3.5.1.5 Ambient light susceptibility (see 4.6.3.6 and 6.6.1). The optical power from any one port or sum of ports shall be not greater than minus 50 decibel milliwatts (dBm).

3.5.2 Mechanical properties. The following mechanical properties shall be met without physical damage to the attenuator.

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3.5.2.1 Cable pigtails. Unless otherwise specified in the specification sheet (see 3.1), attenuators with cable pigtails shall meet the mechanical properties specified in 3.5.2.1.1 through 3.5.2.1.5.

3.5.2.1.1 Impact (see 4.6.4.1.1). The attenuator shall not be visibly damaged, changed in dimensions or show deformations. The attenuator shall meet the optical requirements of 3.5.1.3.

3.5.2.1.2 Twist (see 4.6.4.1.2). Cable-to-attenuator seals shall not be rendered inoperable nor shall any deformation occur. The attenuator shall meet the optical requirements of 3.5.1.3.

3.5.2.1.3 Cable seal flexing (see 4.6.4.1.3). Cable-to-attenuator strain relief mechanisms shall prevent loss of environmental sealing or other damage which may impair the attenuator operation.

3.5.2.1.4 Crush resistance (for package S attenuators only) (see 4.6.4.1.4). The attenuator shall show no evidence of broken parts or accessories, or damage to seals. The attenuator shall meet the optical requirements of 3.5.1.3.

3.5.2.1.5 Cable pull-out force (see 4.6.4.1.5). The minimum cable-to-attenuator pull-out strength shall be 50 percent of the cable strength not to exceed mass of 73.5 kilogram (kg) for splices attaching to the strength members of the cable, or 9 kg for splices attaching directly to fiber cable components. The splices shall show no evidence of cable jacket damage, cable clamp failure, cable-to-splice seal damage, distortion or bending of splice parts or cable disengagement from the clamp. The splice shall meet the optical requirements specified in 3.5.1.3.

3.5.2.2 Fiber pigtails. Unless otherwise specified in the specification sheet (see 3.1), attenuators with fiber pigtails shall meet the mechanical properties specified in 3.5.2.2.1 and 3.5.2.2.2.

3.5.2.2.1 Impact (see 4.6.4.2.1). The attenuator shall not be visibly damaged. The attenuator shall meet the optical requirements of 3.5.1.3.

3.5.2.2.2 Fiber dynamic strength (see 4.6.4.2.2). The minimum fiber dynamic strength shall be 50 percent of the specified proof strength of the attached fiber. There shall be no fiber-to-attenuator seal damage, or distortion or bending of attenuator parts. The attenuator shall meet the optical requirements specified in 3.5.1.3.

3.5.2.3 Connectorized. Unless otherwise specified in the specification sheet (see 3.1), attenuators with connectors as the port construction shall meet the mechanical properties specified in 3.5.2.3.1 through 3.5.2.3.4.

3.5.2.3.1 Impact (see 4.6.4.3.1). The attenuator shall not be visibly damaged, change dimensions or show deformation. The attenuator shall meet the optical requirements of 3.5.1.3.

3.5.2.3.2 Crush resistance (for package S attenuators only) (see 4.6.4.3.2). The attenuator shall show no evidence of broken parts or damage to seals. The attenuator shall meet the optical requirements of 3.5.1.3.

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3.5.2.3.3 Connector pull-out force (see 4.6.4.3.3). The minimum connector-to-attenuator pull-out strength shall be 50 percent the specified tensile strength of the attached connector or a mass of 9 kg, whichever is less. There shall be no evidence of connector-to-attenuator seal damage, distortion or bending of attenuator parts, or connector disengagement from the attenuator. The attenuator shall meet the optical requirements specified in 3.5.1.3.

3.5.2.3.4 Mechanical life (see 4.6.4.3.4). The change in insertion loss after 500 remating cycles shall be not greater than 0.3 dB.

3.5.2.4 Cycling/repeatability of dynamic range (for type V attenuators only) (see 4.6.4.4). The attenuator shall meet the requirements of 3.5.1.3 after 500 cycles from the minimum insertion loss to the maximum insertion loss. The attenuator shall meet the requirements of 3.5.1.3 at both settings.

3.5.3 Environmental properties. The attenuator shall meet all requirements as specified (see 3.1), during the specified operating environments and after the specified storage environment. The operating temperature range and storage temperature range shall be as shown in table I, as specified in the specification sheet (see 3.1).

TABLE I. Temperature ranges.

Range	Operating (°C)	Storage (°C)
1	-54 to +65	-62 to +71
2	-54 to +85	-62 to +85
3	-28 to +65	-62 to +71
4	-28 to +85	-62 to +85

3.5.3.1 Temperature life (see 4.6.5.1). The attenuator shall show no evidence of dimensional change, opening of seals, cracking or crazing of components, or any form of physical damage which will impair performance. The attenuator shall meet the optical requirements of 3.5.1.3. When a hermetic enclosure is specified, the attenuator shall meet the requirements of 3.5.3.6.

3.5.3.2 Thermal shock (see 4.6.5.2). The attenuator shall be fully functional and show no evidence of physical degradation. The attenuator shall meet the optical requirements of 3.5.1.3. When a hermetic enclosure is specified, the attenuator shall meet the requirements of 3.5.3.6.

3.5.3.3 Temperature-humidity cycling (see 4.6.5.3). The attenuator shall show no evidence of deterioration of parts or materials, loosening of finishes, physical distortion, corrosion of metals, entrapment of moisture, separation of bonded surfaces, or physical damage. The attenuator shall meet the optical requirements of 3.5.1.3.

3.5.3.4 Salt spray (corrosion) (see 4.6.5.4). The attenuator shall show no evidence of salt penetration into the attenuator seals nor evidence of corrosive effects, peeling, flaking or color change in materials.

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the cold chamber, the specimen shall be wound around the attached mandrel

3.5.3.5 Water pressure (see 4.6.5.5). The attenuator shall reveal no evidence of the penetration of the indicator dye into the sealed region of the attenuator. The attenuator shall meet the optical requirements of 3.5.1.3.

3.5.3.6 Hermetic (for hermetic enclosures only) (see 4.6.5.6). When a hermetic enclosure is specified, the attenuator shall be permanently sealed to prevent transmission of air, gas, or moisture.

3.5.3.7 Freezing water (see 4.6.5.7). The attenuator shall not be physically damaged during exposure. The attenuator shall meet the optical properties of 3.5.1.3.

3.5.3.8 Fluid immersion (see 4.6.5.8). The attenuator shall reveal no evidence of swelling or softening of material, nor loss of sealing capability or other effects detrimental to the operation of the attenuator.

3.5.3.9 Sand and dust (see 4.6.5.9). The attenuator shall show no evidence of physical degradation. The attenuator shall meet the optical properties of 3.5.1.3.

3.5.3.10 Shock (see 4.6.5.10). The attenuator shall show no evidence of physical damage that impairs its specified function. The attenuator shall meet the optical requirements of 3.5.1.3. When a hermetic enclosure is specified, the attenuator shall meet the requirements of 3.5.3.6.

3.5.3.11 Vibration (see 4.6.5.11). Attenuators shall not split, crack or show deformation that impairs its specified function. The attenuator shall meet the optical requirements of 3.5.1.3. When a hermetic enclosure is specified, the attenuator shall meet the requirements of 3.5.3.6.

3.5.3.12 Nuclear radiation resistance (see 4.6.5.12). The performance requirements of 3.5.1.3 shall be met when tested in accordance with 4.6.5.12.

3.5.4 Chemical properties.

3.5.4.1 Fungus resistance (see 4.6.6.1). The attenuator materials shall be in accordance with MIL-STD-454, requirement 4, for fungus-inert materials or shall meet grade I classification of MIL-STD-810, method 508.

3.5.4.2 Flame survivability (see 4.6.6.2). The attenuator shall not produce explosive fumes or cause any dripping material to ignite. The attenuator shall meet the optical requirements specified in 3.5.1.3.

3.6 Transportability. The attenuator operational performance specified herein shall not be affected following storage or shipment under environmental conditions for shipment and storage specified herein, and as required for packaging for shipment (see 5.1).

3.7 Identification marking (see 4.6.1.1). Unless otherwise specified, the attenuator shall be identified by a marking applied to the outer surface of the attenuator. When practical, the identification marking shall consist of the following:

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(a) Specification sheet part number.
(b) Manufacturer's code designation.
(c) The words "Fiber Optic Attenuator".
(d) Date code (4 digit - week, year).

The characteristics and method of identification marking shall conform to the requirements of MIL-STD-1285. The identification marking shall be permanent and legible in accordance with the marking requirements of MIL-STD-1285. Identification marking shall be applied with the vertical axes of the printed characters perpendicular to the longest dimension of the attenuator housing. Characters shall remain legible for the lifetime of the attenuator.

3.8 Workmanship (see 4.6.1.1). The following are minimal considerations and are not intended to exclude other pertinent workmanship requirements deemed necessary by the contractor:

(a) Outer surface shall be free of rust, burnt areas, abrasions, holes, roughead areas, bulges, thin spots, and indentations.
(b) Inner parts shall be free of fractures, holes, discontinuities, bulges, and thin spots.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

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4.2.1 Toxicological product formulations. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable portions of EIA 455.

4.4 First article inspection. First article inspection shall consist of all the tests listed in table II and as specified in the specification sheet (see 3.1).

TABLE II. First article inspection.

Inspection	Requirement	Test method	Sample size
Group I			
Visual and mechanical	3.4, 3.8	4.6.1.1	8
Finish	3.4.8	4.6.1.1	1/
Size	3.4.9	.4.6.1.1	1/
Weight	3.4.10	4.6.2.1	1/
Color	3.4.11	4.6.1.1	1/
Identification marking	3.7	4.6.1.1	1/
Insertion loss	3.5.1.1.1 3.5.1.2.1	4.6.3.1	1/
Dynamic range	3.5.1.2.2	4.6.3.2	1/
Resolution	3.5.1.2.3	4.6.3.3	1/
Return loss	3.5.1.4	4.6.3.5	1/
Ambient light susceptibility	3.5.1.5	4.6.3.6	1/
Group II			
Water pressure	3.5.3.5	4.6.5.5	1
Hermetic	3.5.3.6	4.6.5.6	1
Flame survivability	3.5.4.2	4.6.6.2	1

See footnotes at end of table.

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TABLE II. First article inspection - Continued.

Inspection	Requirement	Test method	Sample size
Group III			
Temperature life	3.5.3.1	4.6.5.1	1
Thermal shock	3.5.3.2	4.6.5.2	2/
Temperature-humidity cycling	3.5.3.3	4.6.5.3	2/
Salt spray	3.5.3.4	4.6.5.4	2/
Freezing water	3.5.3.7	4.6.5.7	2/
Fluid immersion	3.5.3.8	4.6.5.8	2/
Sand and dust	3.5.3.9	4.6.5.9	1
Shock	3.5.3.10	4.6.5.10	3/
Vibration	3.5.3.11	4.6.5.11	3/
Nuclear radiation resistance	3.5.3.12	4.6.5.12	3
Fungus resistance	3.5.4.1	4.6.6.1	4/
Group IV			
Type A attenuators (cable pigtails)			
Impact	3.5.2.1.1	4.6.4.1.1	1
Twist	3.5.2.1.2	4.6.4.1.2	5/
Cable seal flexing	3.5.2.1.3	4.6.4.1.3	5/
Crush resistance	3.5.2.1.4	4.6.4.1.4	5/
Cable pull-out force	3.5.2.1.5	4.6.4.1.5	5/
Type B attenuators (fiber pigtails)			
Impact	3.5.2.2.1	4.6.4.2.1	1

See footnotes at end of table.

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TABLE II. First article inspection - Continued.

Inspection	Requirement	Test method	Sample size
Fiber dynamic strength	3.5.2.2.2	4.6.4.2.2	5/
Type C attenuator (connectorized)			
Impact	3.5.2.3.1	4.6.4.3.1	1
Crush resistance	3.5.2.3.2	4.6.4.3.2	5/
Connector pull-out force	3.5.2.3.3	4.6.4.3.3	5/
Mechanical life	3.5.2.3.4	4.6.4.3.4	5/
Type A, B, and C attenuators			
Cycling/repeatability	3.5.2.4	4.6.4.4	5/

1/ The same samples shall be used as in the visual and mechanical inspection.
2/ The same sample shall be used as in the temperature life inspection.
3/ The same sample shall be used as in the sand and dust inspection.
4/ One sample from the nuclear radiation inspection shall be used.
5/ The same sample shall be used as in the respective impact inspection.

4.4.1 Sample. An attenuator sample shall be submitted for each attenuator construction for which first article approval is desired. The sample submitted shall be eight attenuators of the same part number.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table II in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall then be divided into the three groups. The sample units shall then be subjected to the inspection for their particular group. After completion of sample testing, all units shall be resubjected to group I testing. Any attenuator failing any inspection shall not be subjected to further inspection.

4.4.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the inspections and tests specified for group A inspection (table III), group B inspection (table IV), and group C inspection (table V) (see 6.5).

4.5.1 Group A inspection. Group A inspections shall follow the order shown in table III.

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TABLE III. Group A inspection.

Inspection	Requirement	Test method
Visual and mechanical	3.4, 3.8	4.6.1.1
Finish	3.4.8	4.6.1.1
Size	3.4.9	4.6.1.1
Weight	3.4.10	4.6.2.1
Color	3.4.11	4.6.1.1
Identification marking	3.7	4.6.1.1
Insertion loss	3.5.1.1.1 3.5.1.2.1	4.6.3.1
Dynamic range	3.5.1.2.2	4.6.3.2
Resolution	3.5.1.2.3	4.6.3.3
Return loss	3.5.1.4	4.6.3.5
Ambient light susceptibility	3.5.1.5	4.6.3.6

4.5.2 Group B inspection. Group B inspection shall consist of the inspections specified in table IV. Group B inspections shall be made on units that have passed the group A inspection.

TABLE IV. Group B inspection.

Inspection	Requirement	Test method
Shock	3.5.3.10	4.6.5.10
Thermal shock	3.5.3.2	4.6.5.2
Type A attenuators (cable pigtails)		
Impact	3.5.2.1.1	4.6.4.1.1
Twist	3.5.2.1.2	4.6.4.1.2
Cable seal flexing	3.5.2.1.3	4.6.4.1.3
Crush resistance	3.5.2.1.4	4.6.4.1.4
Cable pull-out force	3.5.2.1.5	4.6.4.1.5

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TABLE IV. Group B inspection - Continued.

Inspection	Requirement	Test method
Type B attenuators (fiber pigtails)		
Impact	3.5.2.2.1	4.6.4.2.1
Fiber dynamic strength	3.5.2.2.2	4.6.4.2.2
Type C attenuators (connectorized)		
Impact	3.5.2.3.1	4.6.4.3.1
Crush resistance	3.5.2.3.2	4.6.4.3.2
Connector pull-out force	3.5.2.3.3	4.6.4.3.3

4.5.3 Group C inspections. Group C inspections shall consist of the inspections in table V. Group C inspection shall have satisfactorily completed all group A and group B inspections. After completion of group C inspections, the samples shall be re-subjected to group A inspection.

4.5.3.1 Disposition of group C inspections sample units. Sample units subjected to group C inspection that have failed shall not be delivered.

TABLE V. Group C inspection.

Inspection	Requirement	Test method
Water pressure	3.5.3.5	4.6.5.5
Hermetic	3.5.3.6	4.6.5.6
Flame survivability	3.5.4.2	4.6.6.2
Temperature life	3.5.3.1	4.6.5.1
Temperature-humidity cycling	3.5.3.3	4.6.5.3
Salt spray	3.5.3.4	4.6.5.4
Freezing water	3.5.3.7	4.6.5.7
Fluid immersion	3.5.3.8	4.6.5.8
Sand and dust	3.5.3.9	4.6.5.9

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TABLE V. Group C inspection - Continued.

Inspection	Requirement	Test method
Vibration	3.5.3.11	4.6.5.11
Cycling/repeatability	3.5.2.4	4.6.4.4
Nuclear radiation resistance	3.5.3.12	4.6.5.12
Fungus resistance	3.5.4.1	4.6.6.1
Mechanical life	3.5.2.3.4	4.6.4.3.4

4.6 Methods of inspection.

4.6.1 General testing practice.

4.6.1.1 Visual and mechanical inspections. Unless otherwise specified (see 3.1), visual and mechanical examinations shall be performed in accordance with EIA 455-13 to verify that the design, construction, physical characteristics, dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.6.1.2 Optical test methods. The use of optical test methods to assess the influence of test exposures on performance of attenuators requires that they be connected to lengths of fiber. Unless otherwise specified (see 3.1), the optical properties inspections shall be performed at a wavelength of 1.310 ± 0.020 micrometers (μm). The attenuators shall be attached to lengths of cable of the type specified in MIL-C-85045. The cable lengths shall be not greater than 30 meters. The optical fiber components exposed at the test set (source, detector, or coupler) may be connectorized for access to the test set. For multimode fiber, the source used shall be noncoherent. Light launch conditions shall be in accordance with table VI.

TABLE VI. Light launch conditions.

Fiber type	Launch condition
Single-mode	30-mm diameter mandrel
Multimode	70/100 restricted, or 70/70 restricted (see 6.6.5 and 6.6.6)

Any optical power detection method may be used if the method is sufficiently sensitive to measure the differential optical power levels as specified (see 3.5.1.1), and if the method provides repeatable readings (less than 3 percent variation).

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4.6.1.3 Variable attenuators. Unless otherwise specified (see 3.1), type V attenuators shall be set at mid-range of the specified range of optical attenuation for the environmental and mechanical tests.

4.6.1.4 Equivalent test methods. The use of equivalent test methods is allowed (see 3.1 and 6.2).

4.6.2 Construction inspection.

4.6.2.1 Weight (see 3.4.10). The attenuator weight shall be tested using scales of sufficient range and accuracy appropriate for the specified weight and tolerance.

4.6.2.2 Size (see 3.4.9). Attenuators shall be measured using instruments with accuracies such that systematic errors are within the tolerances defined in the specification sheet (see 3.1).

4.6.3 Optical properties inspection.

4.6.3.1 Insertion loss (see 3.5.1.1.1 and 3.5.1.2.1). The insertion loss shall be measured in accordance with EIA 455-34.

4.6.3.2 Dynamic range (for type V attenuators only) (see 3.5.1.2.2). The attenuator shall be tested in accordance with 4.6.3.1 at the maximum and minimum levels of insertion loss.

4.6.3.3 Resolution (see 3.5.1.2.3). The attenuator shall be tested in accordance with 4.6.3.1 for resolution of each of the specified attenuator steps.

4.6.3.4 Change in optical transmittance (see 3.5.1.3). The change in optical transmittance shall be measured in accordance with EIA 455-20 or equivalent (see 4.6.1.4), using a monitor fiber.

4.6.3.5 Return loss (see 3.5.1.4). The return loss shall be measured in accordance with appendix A (see 6.3 and appendix B) of this specification.

4.6.3.6 Ambient light susceptibility (see 3.5.1.5). The ambient light susceptibility shall be measured in accordance with EIA 455-22, with the exception that the output power in the "on" state shall be referenced to 1 milliwatt. The test temperature shall be 25 degrees Celsius (°C).

4.6.4 Mechanical properties inspection.

4.6.4.1 Cable pigtails.

4.6.4.1.1 Impact (see 3.5.2.1.1). The attenuator shall be tested in accordance with EIA 455-2. Package S attenuators shall be classified as moderate service class, and package P attenuators shall be classified as light service

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class. The change in optical transmittance shall be measured after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

4.6.4.1.2 Twist (see 3.5.2.1.2). The attenuator shall be tested in accordance with EIA 455-36, for 50 cycles, with the exception that the test sample shall be a properly pigtailed attenuator. The tensile load shall be 5 newtons. The change in optical transmittance shall be measured during and after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined with 4.6.1.1.

4.6.4.1.3 Cable seal flexing (see 3.5.2.1.3). The attenuator shall be tested in accordance with MIL-STD-1344, method 2017, with the exception that the test sample shall be a properly pigtailed attenuator. At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

4.6.4.1.4 Crush resistance (for package S attenuators only) (see 3.5.2.1.4). The attenuator shall be tested in accordance with MIL-STD-1344, method 2008, with the exception that the test sample shall be a properly pigtailed attenuator. The test load shall be 1250 newtons and the number of loads shall be one. The change in optical transmittance shall be measured during and after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

4.6.4.1.5 Cable pull-out force (see 3.5.2.1.5). The attenuator shall be tested in accordance with EIA 455-6. The axial tensile load shall be applied to the load specified and shall be maintained for 10 minutes. The change in optical transmittance shall be measured during and after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

4.6.4.2 Fiber pigtails.

4.6.4.2.1 Impact (see 3.5.2.2.1). The attenuator shall be tested in accordance with EIA 455-2, with the exception that the test sample shall be an attenuator with fiber pigtails. Attenuators with fiber pigtails shall be classified as light service class. The change in optical transmittance shall be measured after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

4.6.4.2.2 Fiber dynamic strength (see 3.5.2.2.2). The fiber pigtail dynamic strength shall be tested as follows: The fiber pigtail shall have an axial tensile load applied up to the load specified at an angle of 45 degrees to the normal (see figure 1). With the load applied, the pigtail shall be rotated through one rotation (360 degrees). The change in optical transmittance shall be measured during and after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

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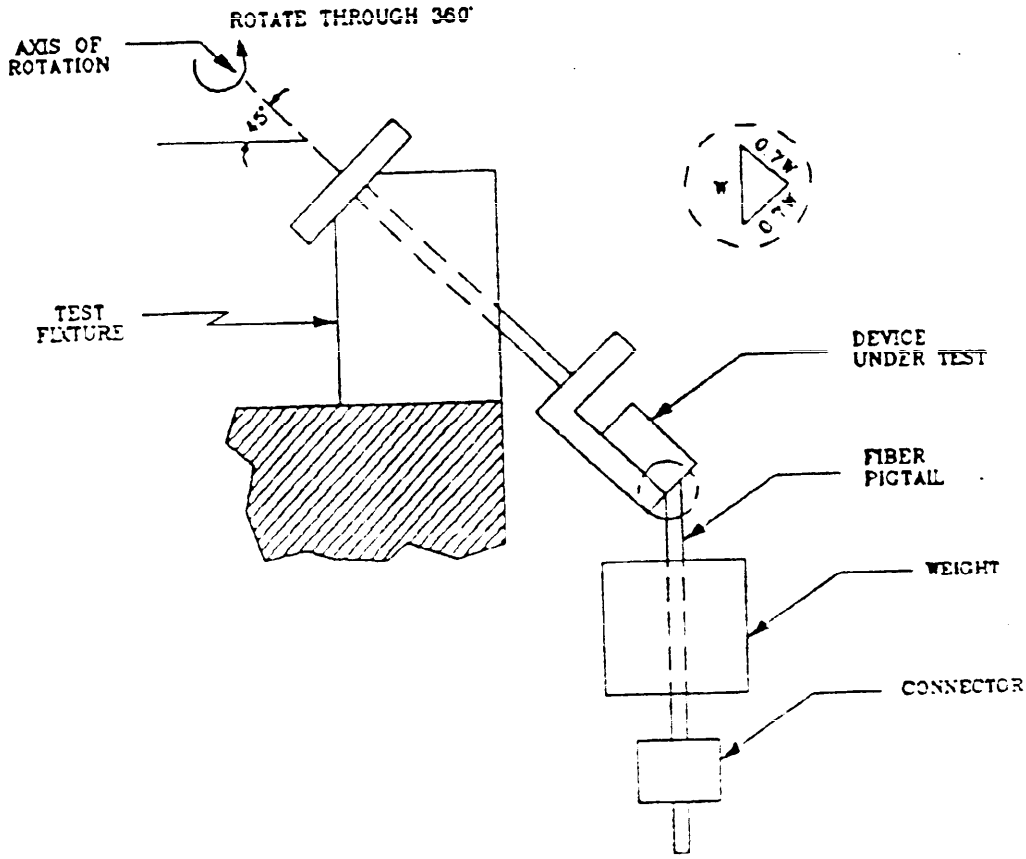


FIGURE 1. Fiber dynamic strength test setup.

4.6.4.3 Connectorized.

4.6.4.3.1 Impact (see 3.5.2.3.1). The attenuator shall be tested in accordance with EIA 455-2. Package S attenuators shall be classified as moderate service class, and package P attenuators shall be classified as light service class. The change in optical transmittance shall be measured after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

4.6.4.3.2 Crush resistance (for package S attenuators only) (see 3.5.2.3.2). The attenuator shall be tested in accordance with MIL-STD-1344, method 2008, with the exception that the test sample shall be an attenuator with connectors as the port construction. The test load shall be 1250 newtons, and the number of loads shall be one. The change in optical transmittance shall be measured during and after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

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4.6.4.3.3 Connector pull-out force (see 3.5.2.3.3). The attenuator shall be tested in accordance with EIA 455-6, with the exception that the test sample shall be a connector. The axial tensile load shall be applied to the load specified and shall be maintained for 10 minutes. The change in optical transmittance shall be measured during and after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

4.6.4.3.4 Mechanical life (see 3.5.2.3.4). The attenuator shall be tested for optical transmittance after 500 remating cycles in accordance with EIA 455-21. The change in optical transmittance shall be measured for every 50 cycles (see 4.6.3.4).

4.6.4.4 Cycling/repeatability (for type V attenuators only) (see 3.5.2.4). The attenuator shall be tested for optical transmittance for 500 cycles from the minimum insertion loss to the maximum insertion loss. The attenuator shall be set at the minimum and maximum settings, and the optical transmittance shall be measured at both settings for every 50 cycles.

4.6.5 Environmental properties inspection.

4.6.5.1 Temperature life (see 3.5.3.1). The attenuator shall be tested in accordance with EIA 455-4. The exposure time shall be 96 hours. For package S attenuators, the exposure temperature shall be 105°C. For package P attenuators, the exposure temperature shall be 125°C. The change in optical transmittance shall be measured after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1 and tested for hermeticity (see 4.6.5.6), if applicable to the attenuator design.

4.6.5.2 Thermal shock (see 3.5.3.2). The attenuator shall be tested for 10 cycles in accordance with EIA 455-71 schedule C. The high and low temperatures shall correspond to the specified storage temperature extremes. The change in optical transmittance shall be measured after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1 and tested for hermeticity (see 4.6.5.6), if applicable to the attenuator design.

4.6.5.3 Temperature-humidity cycling (see 3.5.3.3). The attenuator shall be tested in accordance with EIA-455-5. The relative humidity shall be 95 percent and the temperature range shall be the specified operating temperature range. The change in optical transmittance shall be measured during and after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

4.6.5.4 Salt spray (corrosion) (see 3.5.3.4). The attenuator shall be tested in accordance with EIA 455-16. The exposure time shall be 96 hours, and the exposure temperature shall be 35°C. At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

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4.6.5.5 Water pressure (see 3.5.3.5). The attenuator shall be immersed in an aqueous dye penetrant solution at an applied pressure of 0.1 megapascal (equivalent to a depth of 10.4 meters) for 48 hours at a temperature of 10 to 35°C. The change in optical transmittance shall be measured after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

4.6.5.6 Hermetic (see 3.5.3.6). The attenuator shall be tested for hermetic seal in accordance with MIL-STD-883, method 1014. Test condition A1 shall be used for the seal test, and test condition C shall be used for the gross leak test.

4.6.5.7 Freezing water (see 3.5.3.7). The attenuator shall be tested in accordance with DOD-STD-1678, method 4050. The attenuator shall be completely immersed in water for 6 hours at the specified minimum operating temperature. The vessel containing the water need not be sealed. The size (volume) of the vessel shall be approximately two to three times the size of the attenuator under test. The change in optical transmittance shall be measured during and after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

4.6.5.8 Fluid immersion (see 3.5.3.8). The attenuator shall be tested in accordance with EIA 455-12. Each attenuator shall be immersed in each immersion fluid as specified in table VII for a period of 24 hours. Each attenuator shall be completely dried before further testing. At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

TABLE VII. Immersion test fluids.

Fluids	Specification	Test temperature (°C)
Fuel oil	MIL-F-16884	33 - 37
Turbine fuel (JP-5)	MIL-T-5624	20 - 25
Isopropyl alcohol	TT-I-735	20 - 25
Hydraulic fluids	MIL-H-5606	48 - 50
Lubricating oils	MIL-L-17331 MIL-L-23699	73 - 77
Sea water	-	20 - 25

4.6.5.9 Sand and dust (see 3.5.3.9). The attenuator shall be tested in accordance with EIA 455-35. The change in optical transmittance shall be measured during and after the test (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1.

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4.6.5.10 Shock (see 3.5.3.10). The attenuator shall be tested in accordance with MIL-S-901, grade A, type A, class I. Package P attenuators shall be tested in accordance with MIL-STD-883, method 2002, test condition B. Unless otherwise specified in the specification sheet (see 3.1), the change in optical transmittance shall be monitored continuously during and after the test with equipment having a time resolution of at least 50 microseconds (μs) (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1 and tested for hermeticity (see 4.6.5.6), if applicable to the attenuator design.

4.6.5.11 Vibration (see 3.5.3.11). The attenuator shall be tested in accordance with MIL-STD-1344, method 2005, test conditions II and VI. For test condition VI, test condition C shall apply. Test condition I shall be used during the low frequency portion of test condition II. Unless otherwise specified in the specification sheet (see 3.1), the change in optical transmittance shall be monitored continuously during and after the test with equipment having a time resolution of at least 50 μs (see 4.6.3.4). At the completion of the test, the attenuator shall be visually examined in accordance with 4.6.1.1 and tested for hermeticity (see 4.6.5.6), if applicable to the attenuator design.

4.6.5.12 Nuclear radiation resistance (see 3.5.3.12). The attenuator shall be tested in accordance with EIA 455-49. Unless otherwise specified in the specification sheet (see 3.1), tests shall be performed at a wavelength of 1.310 ± 0.020 μm and at the low operating temperature, 20°C, and the high operating temperature. The dose rate shall be 30 rads per second. The total dose shall be as specified (see 6.2). The change in optical transmittance shall be measured during and after the test (see 4.6.3.4).

4.6.6 Chemical properties inspection.

4.6.6.1 Fungus resistance (see 3.5.4.1). Attenuators that do not meet the requirements of fungus-inert materials in accordance with MIL-STD-454, requirement 4, shall be tested for resistance to fungus in accordance with MIL-STD-810, method 508.

4.6.6.2 Flame survivability (see 3.5.4.2). The attenuator shall be tested in accordance with MIL-STD-1344, method 1012, test condition A, with the exception that the test sample shall be an attenuator instead of a connector. The change in optical transmittance shall be measured after the test (see 4.6.3.4).

4.7.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-55330. The level of packing shall be as specified (see 6.2).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The attenuators covered by this specification are intended for use in any application where their performance characteristics are required. The attenuators are for installation in shipboard systems within the limitations of applicable performance requirements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of the specification.
(b) Quantity required.
(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Applicable specification sheet number, title, and date (see 3.1).
(e) When first article inspection is required (see 3.2).
(f) Equivalent test methods (see 4.6.1.4).
(g) Total dose radiation requirement (see 4.6.5.12).
(h) Level of packaging required (see 5.1).
(i) Applicable part number (see 6.8).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.6.3.5 and appendix B	DI-MISC-80653	Test reports	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ___ production items, a standard production item from the contractor's current inventory (see 3.2), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles.

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Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Quality conformance. Quality conformance inspections require contractual definition of the overall test program, including sample sizes and lot sizes, if appropriate (see 4.5).

6.6 Definitions.

6.6.1 Ambient light susceptibility. Ambient light susceptibility is the optical power (dBm) that enters the attenuator from ambient illumination incident upon the attenuator.

6.6.2 Fiber optic attenuator. A device that operates upon its input optical signal power level in such a way that its output signal power level is less than the input level.

6.6.3 Insertion loss. Insertion loss is the radiant power loss (dB) caused by absorption, scattering, diffusion, leaky waves, dispersion, microbends, macrobends, reflection, radiation, or other causes when an attenuator is inserted into the system.

6.6.4 Return loss. Return loss is the optical power (dB) that is reflected back toward the source of optical power by the attenuator.

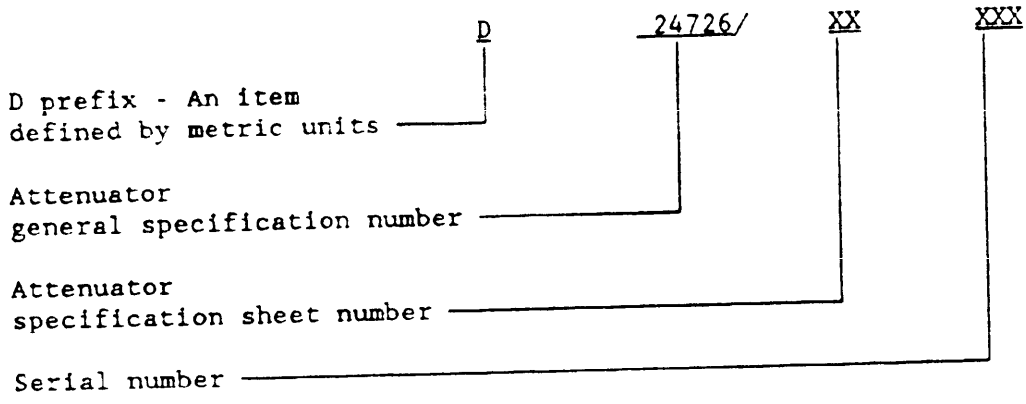
6.6.5 70/70 restricted launch. A 70/70 restricted launch is a beam optics launch with a 70 percent spot size and source aperture equal to 70 percent of the fiber numerical aperture.

6.6.6 70/100 restricted launch. A 70/100 restricted launch is a beam optics launch with a 70 percent spot size and source aperture equal to or greater than the fiber numerical aperture.

6.7 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

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6.8 Part or Identifying Number (PIN). Attenuators should contain only the following:



Examples: D24726/01-001
D24726/02-002

6.9 Part designator. A part designator, if specified (see 3.1), should include classifications (see 1.2) as follows:

(a) Type (see 1.2.1).
(b) Port construction (see 1.2.2).
(c) Package (see 1.2.3).

Example: FBS.

6.10 Subject term (key word) listing.

Change in optical transmittance
Fiber optic communications
Fiber optic component
Fixed attenuator
Insertion loss
Variable attenuator

Review activities:
Navy - EC, YD

Preparing activity:
Navy - SH
(Project 6006-N001)

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WITHOUT THE TAPE. THE TAPE SHALL BE 0.25 (6.3) MM NOMINAL OF WIDTH. THE

APPENDIX A

TEST PROCEDURE TO MEASURE RETURN LOSS

10. SCOPE

10.1 Scope. This procedure determines the ratio of optical power reflected by a component or an assembly to the optical power incident on a component port when that component assembly is introduced into a link or system. This ratio is termed return loss. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

10.2 Test. The test for return loss is based on two setups. The first test setup shown on figure 2 is designed to measure the insertion loss of a coupler along a particular through path of that device. The coupler is an integral part of measuring the insertion loss of the test setup shown on figure 3.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. TEST EQUIPMENT

30.1 Apparatus. The apparatus shall consist of the following primary items and those additional items identified on figures 2 and 3 and the specification sheet. The apparatus configuration shall be as shown on figures 2 and 3.

30.2 Optical power source. The optical power source shall consist of a light source emitting wavelengths suitable for the devices tested and can either be modulated or unmodulated. A light-emitting diode (LED) or injection laser with suitable radiation band is a suitable source and may be specified. The source shall be stable to plus or minus 1 percent (0.05 dB) over the measurement period. Wavelength selective filters may be used if necessary to limit optical spectral width.

30.3 Optical directional coupler. The directional coupler (or splitter) may be of any convenient coupling ratio. Its return path to forward path power ratio (directivity) when terminated in a nonreflecting termination shall be 10 dB greater than the maximum return loss ratio desired to be measured. The coupler shall be mode insensitive.

30.4 Test jumper assemblies. One pair of pigtails are required per device under test (DUT) measurement.

30.5 Termination. An optical load which is fully absorbent and nonreflective (at least 10 dB below the desired level of test sensitivity) when coupled to the connecting cable of the test system. Index matching gel or fluid is a suitable termination.

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30.6 Optical power meter. A device capable of measuring incident optical power which is linear to within 5 percent over the power levels to be measured and stable to plus or minus 1 percent (0.05 dB) over the measurement period. Relative power levels only are required so no absolute accuracy is necessary.

30.7 Interconnects/splices. Since all connectors used in the test will influence the test results, extremely low loss and repeatable units shall be used. Fusion splices are recommended if practical and are noted as such on the figures. Index matching materials at all interfaces not under test are highly recommended.

30.8 Mode filter. A 50-millimeter (mm) diameter loop of fiber/cable of the same type used in the coupler.

40. TEST SAMPLE

40.1 Test sample. The test sample or DUT shall be as specified (see 3.1).

50. TEST PROCEDURE

50.1 Determination of insertion loss of coupler. The test procedure shall be as follows:

(a) Referring to the test configuration shown in step A of figure 2, a meter of single-mode fiber is fusion-welded to the lead length of the fiber from port 3 of the coupler as a temporary joint. This fiber lead from port 2 is connected to a detector/power meter. Index matching material is applied to fiber leads from ports 1 and 4 to provide termination points. A power measurement is taken at the specified wavelength and a measurement P_2 is recorded.

(b) In step B, cut the fiber between the coupler and the fusion joint, just beyond the joint, allowing material for polishing of the cut end. The free end of the fiber shall be properly prepared so that it is smooth and perpendicular to the axis. Connect the prepared end of the fiber to the detector ensuring that the power level is repeatable. Measure and record this power measurement as P_3. The insertion loss of this particular path between port 2 and port 3 of the coupler is given by the formula:

$$L_c = -10\log_{10} (P_2/P_3)$$

50.2 Determination of DUT return loss. The test procedure shall be as follows:

(a) Using the setup shown on figure 3, connect the fiber lead of port 1 of the coupler to the light source. Connect the detector/power meter to the fiber lead from port 3. Terminate the fiber leads from port 2 and 4 with index matching gel. Measure and record the insertion loss between ports 1 and 3 as P_o.

(b) Switch the detector/power meter to the fiber lead from port 2 of the coupler and terminate the fiber from port 3 with index gel. Measure and record the insertion loss between ports 1 and 2 as P_1.

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(c) The test jumper assemblies are cut in half to provide the necessary DUT pigtails required for the return loss measurements. Joint the DUT pigtail to the fiber from port 3 with a temporary joint (fusion). The DUT is then coupled to a second pigtail using an adapter. The end of the second pigtail is terminated with index gel. With the light source at the specified wavelength, measure and record the power reflection of the DUT as P'_i. Repeat steps A through C for the remaining DUT's to be measured.

(d) The return loss of each DUT is then calculated by the formula below:

$$R_L = -10\log_{10} (P'_i - P_i)/P_o + 10\log_{10} (P_2/P_3)$$

The DUT should have (as an objective) a return loss greater than 30 dB.

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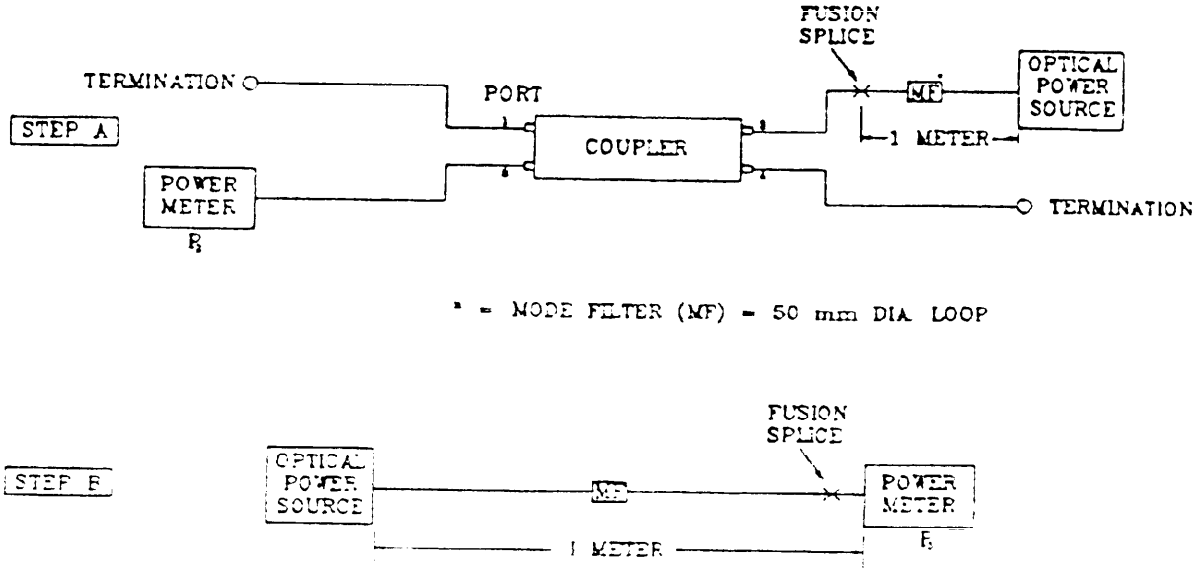


FIGURE 2. Coupler insertion loss test setup.

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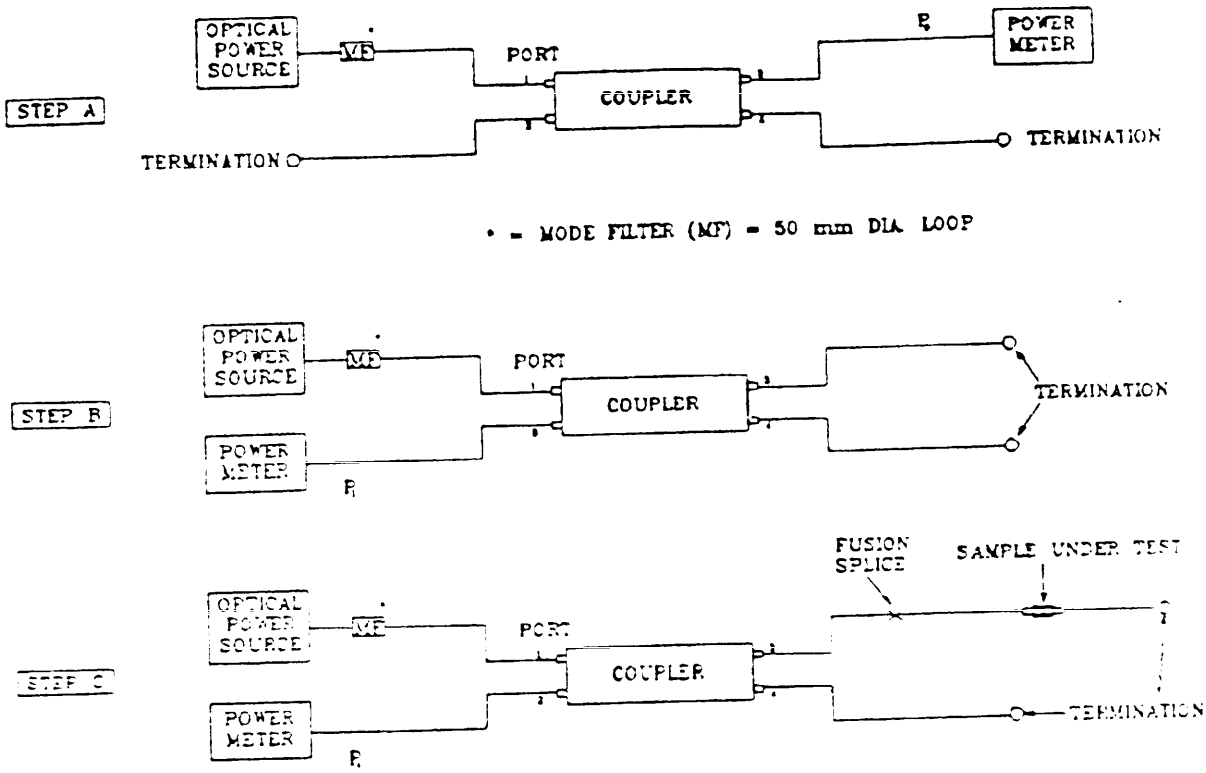


FIGURE 3. DUT return loss test setup.

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APPENDIX B

TEST REPORTS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that shall be included on test reports when required by the contract or order. This appendix is mandatory only when data item description DI-MISC-80653 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. TEST REPORTS

30.1 Return loss (see 4.6.3.5). When required by the contract or order, return loss test reports shall include the following information:

(a) Test title.
(b) Test sample description.
(c) Test equipment used and calibration.
(d) Launch conditions.
(e) Test values and results.
(f) Specification sheet reference.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-24726(NAVY)	ATTENUATORS, FIBER OPTIC, SHIPBOARD GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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